

No need to submit. Please attempt all these questions yourself and reach out for any difficulties. Most of these questions are from [2] and [1] (available here)

- Suppose A and B are 4×2 and 2×4 matrices respectively, such that $AB = 0$ (i.e. AB is the all zero matrix). What are the possible values of $\text{rank}(A)$ and $\text{rank}(B)$? For each possible value of the ranks, construct a suitable example of A and B with those ranks such that $AB = 0$.
- (Optional - requires an additional lecture to complete) Answer if each of the following are true or false, with a brief justification (either a proof or a counter-example). Do not assume anything about the sizes of A and B , other than what is stated and whatever is required for the expression to make sense.
 - If $A^2 = 0$ then $A = 0$.
 - If A is square full rank, then so is A^2 .
 - If A is fat full rank, then $A^T A$ is invertible.
 - If A and B are skinny full rank, so is $[A \ B]$.
 - If A and B are full rank, so is $\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$.
- Consider the following set of vectors in $\mathbb{R}^2$

$$\mathcal{B} = \left\{ \underline{x} \in \mathbb{R}^2 : \|\underline{x}\|_2 = 3 \right\}.$$

You can use the geometric structure of norm balls to answer the questions below.

- Sketch the set of points $\mathcal{B}$.
 - Identify vectors in $\mathcal{B}$ that have the smallest 1-norm, 2-norm and ∞ -norm, respectively.
 - Identify vectors in $\mathcal{B}$ that have the largest 1-norm, 2-norm and ∞ -norm, respectively.
- Consider the matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$
 - Find the nullity and rank of A , and similarly the nullity and rank of A^T (do not use determinants!). It might help to first characterize the null space of A .
 - Show that any for vector in the image/column space of A^T , the sum of all its entries is zero.
 - Suppose $\underline{b}$ is a vector whose entries sum to 0, then does $A^T \underline{x} = \underline{b}$ have a solution?
 - We say that a vector $\underline{x} \in \mathbb{R}^n$ is a *probability vector* if all the entries in $\underline{x}$ are non-negative (i.e. $x_i \geq 0$ for all i) and the entries sum to 1 (i.e. $x_1 + x_2 + \dots + x_n = 1$).
 - Let $\mathcal{P}_n$ denote the set of all probability vectors in $\mathbb{R}^n$. Sketch $\mathcal{P}_2$. In general, is $\mathcal{P}_n$ a subspace of $\mathbb{R}^n$?
 - We are interested in $n \times n$ matrices P such that

$$P\underline{x} \text{ is a probability vector whenever } \underline{x} \text{ is.} \tag{1}$$

Show that P satisfies (1) if and only if all the columns of P are probability vectors.

- Suppose you are given a matrix P and asked to *verify* if it satisfies (1). How many computations (additions, multiplications and comparisons) are required to complete this verification?
- Consider the case $n = 3$. Suppose that P satisfies (1), and each column of P has exactly two non zero entries. What is the minimum and maximum possible rank of P ?

(e) (Optional) Generalize (d) above to an arbitrary n .

6. Suppose $\lambda_i > 0$ are positive reals, and $\alpha_1^2 + \alpha_2^2 + \dots = 1$. Show that

$$\left(\lambda_1 \alpha_1^2 + \lambda_2 \alpha_2^2 + \lambda_3 \alpha_3^2 + \dots\right) \left(\frac{\alpha_1^2}{\lambda_1} + \frac{\alpha_2^2}{\lambda_2} + \frac{\alpha_3^2}{\lambda_3} + \dots\right) \geq 1.$$

When do we have equality above ?

7. What is the maximum value of $x + 2y + 3z$ over all the points in the unit sphere (i.e. over all points (x, y, z) that satisfy $x^2 + y^2 + z^2 = 1$?

8. Suppose a_i, b_i for $i = 1, 2, \dots, n$ are positive reals such that $\sum_i a_i = \sum_i b_i = 1$. Then show that

$$\sum_i \frac{a_i^2}{a_i + b_i} \geq \frac{1}{2}?$$

When do we have equality ?

9. Given a signal $x[n]$ for $n = 1, 2, \dots, N$, consider the correlation defined by

$$R_x[m] = \sum_n x[n]x[n-m].$$

(For this expression to be well defined, you can assume $x[k] = 0$ if $k < 0$)

(a) Given an interpretation of $R_x[m]$ as an inner product. When is the value of $R_x[m]$ large ?

(b) Suppose $R_x[10]$ is almost as large as $R_x[0]$. What does this imply ?

Now consider a signal $x[n]$ for $n = 0, 1, 2, \dots$ that is supposed to be communicated to a destination. The communication is imperfect, however, and we obtained a noisy delayed copy of the signal at the receiver (denoted y) as

$$y[n] = \alpha x[n-M] + \eta[n],$$

where the delay M , the channel gain α are unknown. The noise term $\eta[n]$ is also unknown. To estimate M , we first send a pilot signal x (known to both the transmitter and receiver), and apply the following algorithm: we first compute

$$R_{xy}[m] = \sum_n y[n]x[n-m],$$

and find the value of m for which $R_{xy}[m]$ is the highest. This particular value of m is the delay M .

(a) Briefly explain (in words) why you expect/do not expect this algorithm to work. You can ignore the noise term (or assume $\eta[n] = 0$) if required.

10. Suppose Q is an $n \times n$ orthogonal matrix with entries q_{ij} . For this question, you may find the definition of orthogonality and the Cauchy Schwarz inequality helpful.

(a) Prove that $\sum_{j=1}^n |q_{ij}| \leq \sqrt{n}$ (the left-hand-side is the sum of all absolute values from row i).

Let μ_Q denote the largest entry (in absolute value) of Q :

$$\mu_Q = \max_{i,j} |q_{ij}|$$

(b) Explain why $\mu_Q \leq 1$.

(c) Prove that $\frac{1}{\sqrt{n}} \leq \mu_Q$.

(d) If Q_1 and Q_2 are orthogonal matrices, what is the relationship or inequality satisfied by $\mu_{Q_1 Q_2}$, μ_{Q_1} and μ_{Q_2} ?

Let us say that an $n \times n$ orthogonal matrix Q is *extremal* if $\mu_Q = \frac{1}{\sqrt{n}}$.

- (d) Argue that all the entries of an extremal orthogonal matrix are either $1/\sqrt{n}$ or $-1/\sqrt{n}$.
- (e) Construct an extremal 2×2 , and an extremal 3×3 orthogonal matrix.
- (f) If Q_1 and Q_2 are extremal orthogonal matrices, is $Q_1 Q_2$ also extremal ?
- (g) *Optional* (This part may need some more progress the Linear Algebra Course) Suppose Q is $n \times n$ extremal orthogonal, and suppose it has a $p \times q$ submatrix that has all entries equal to $1/\sqrt{n}$. Then show that $pq \leq n$.

References

- [1] Stephen Boyd and Lieven Vandenberghe. *Introduction to applied linear algebra: vectors, matrices, and least squares*. 2018.
- [2] Ian Stewart and David Tall. *The foundations of mathematics*. OUP Oxford, 2015.